

Lucas Ho

lucas.t.ho.lh@gmail.com | 870-897-4995 | GitHub | LinkedIn

Systems-oriented software engineer with experience in high-performance runtime design, defense-in-depth security architecture, and low-latency concurrent systems in Rust.

TECHNICAL SKILLS

Languages: Rust, Python, C/C++, JavaScript, SQL, Java, Swift

Frameworks: Axum, Tokio, Wasmtime, FastAPI, React, Node.js, Actix-Web

Databases/Tools: Postgres, MongoDB, MySQL, Redis, Git, GitHub, Docker, Linux, Prometheus, OPA

Others: WebAssembly/WASI, seccomp BPF, Async/Await, Concurrency, Lock-Free Architecture, Ed25519, APIs, CI/CD

EDUCATION

University of Oklahoma

Master of Science in Computer Science

GPA: 4.0

Norman, OK

August 2024 – May 2026

University of Oklahoma

Bachelor of Science in Computer Science

GPA: 3.78

Norman, OK

Jan 2023 – May 2025

PROFESSIONAL EXPERIENCE

Applied Research Laboratory

Machine Learning Research Intern

Austin, TX

June 2025 – August 2025

- Automated object detection in Power Spectral Density (PSD) data for a U.S. Navy project by developing a custom deep learning architecture that reduced manual analysis time.
- Achieved a reproduction error loss of less than **0.01%** on clean PSD data by implementing advanced data preprocessing techniques and optimizing model hyperparameters.
- Facilitated technical decision-making among senior scientists by presenting a final model and comprehensive performance analysis effectively communicating complex research findings.

University of Oklahoma

Teaching Assistant, Computer Organization

Norman, OK

Jan 2025 – May 2025

- Improved undergraduate understanding of computer architecture by providing technical support on memory design and I/O organization during one-on-one and group tutoring sessions.
- Enhanced student performance and learning outcomes by delivering timely, constructive feedback through the grading of assignments, quizzes, and exams.

SELECTED PROJECTS

Isolator-V: WASM Sandbox Execution Engine | Rust, Wasmtime, Tokio, github.com/lucastimho/isolator-v-wasm-sandbox
Axum, Next.js, TypeScript, Automerge CRDT, xterm.js

- Sustained **9,600+ req/s** at **P99 under 20 ms** on a 50-slot Wasmtime pool by sampling metrics into a lock-free VecDeque snapshot every 500 ms via tokio::spawn.
- Engineered a **5-layer defense-in-depth** around untrusted WASM, capping each job at 50 MB memory and 50 ms CPU via Wasmtime epochs, kernel syscall filtering, and an in-memory VFS with zero host FS exposure.
- Shipped a real-time execution console with **sub-20 ms cold starts**, streaming stdout/stderr through one Axum WebSocket into an xterm.js WebGL terminal, with Automerge CRDT syncing the file tree across tabs.

APEX-Pay: Agentic Transaction Gateway | Python 3.13, FastAPI, SQLAlchemy 2.0, PostgreSQL, Redis 7, HashiCorp Vault, OPA, Ed25519 github.com/lucastimho/apex-pay

- Mitigated the **69.4% prompt-injection attack-success-rate** documented in ClawSafety by architecting a fail-closed FastAPI gateway minting 60-second scoped credentials via HashiCorp Vault response-wrapping and signing receipts with Ed25519.
- Engineered reliability validated by **248 pytest-asyncio tests** hitting **80 ms P50 / 350 ms P99** on /execute, wrapping the Vault hot path in a three-state circuit breaker with Pydantic Logfire spans.
- Shipped real-time observability streaming **200 events/min** over SSE through a single-broker PostgreSQL LISTEN/NOTIFY plus an atomic Redis Lua sliding-window enforcing \$100/hour per-agent caps, scaling via opt-in monthly partitioning.

Sentient-Cache: Edge-Native Semantic Memory Controller for AI Agents | TypeScript, Bun, Hono, Next.js 15, React 19, Three.js, pgvector, BullMQ, Redis, OPA, mTLS 1.3 github.com/lucastimho/sentient-cache

- Engineered refresh-ahead prefetch keeping ingest response under **140 ms** against a 150 ms simulated embedder by firing top-5 nearest-neighbor pgvector pulls as non-blocking side-effects on every (session_id, task) transition.
- Built fail-closed admission emitting **503** + exponential Retry-After above a **100 MB** queue depth by chaining a token-bucket ResourceSentinel, an Open Policy Agent capability evaluator, and mTLS 1.3 CA-pinned sync.
- Shipped a Next.js 15 + React 19 HUD performing client-side semantic search via a **384-dim feature-hashed embedder**, visualizing cache state through Three.js Points with raycasted picking and an additive-blend GLSL shader.

Lethon-OS: Self-governing memory layer for long-horizon LLM agents | *Python 3.12, github.com/lucastimho/lethon-os*
NumPy, Pydantic v2, Redis, Qdrant, SQLite, Ed25519, Next.js 15, React 19, D3

- Engineered an at-most-one async memory pruner verified by **120 offline tests in under 1 second** by guarding execution with a Redis SET NX lock and applying NumPy-vectorized $U(m, t) = \alpha R + \beta C - \gamma D$ scoring over a K=64 moving reference window.
- Designed an Ed25519 chained-receipt audit pipeline with BLAKE2b-256 Merkle roots and a **7-rule regex MemoryScrubber** by signing receipts through the cryptography library and detecting public-key collisions in a rotation-aware KeyRegistry.
- Shipped a Next.js 15 dashboard compiling to **75.3 kB initial / 177 kB First Load JS** by rendering memory shards in a D3 force-directed graph with utility-encoded radius, brightness, and centripetal positioning.

Hacklahoma: Peak Performance

Best Beginners Winner

- Built a functional iPad AI coach within 24 hours by architecting a multi-modal pipeline for real-time video/voice metrics and mentoring 4 novice hackers.
- Maintained a responsive user interface while running concurrent audio and vision models by optimizing asynchronous workloads using Swift Actors and Async/Await.

NBA Player Archetype Clustering System

- Optimized clustering performance by engineering K-Means and Agglomerative algorithms from scratch using NumPy and vectorized dot-product broadcasting.
- Defined unique player archetypes across 19+ metrics by applying PCA and StandardScaler within a greedy trait-matching mapping system.
- Validated model quality against industry baselines by building a Streamlit dashboard that benchmarked Silhouette and Davies-Bouldin metrics.

HackUTD XI Project: ToyoTrends

Second Place Winner

- Delivered a multi-vehicle carbon emission analysis tool in under 24 hours by integrating CarsXE and SambaNova AI APIs into a React.js web application.
- Generated actionable environmental insights by implementing dynamic data visualizations using the Chart.js library.